Transfer: thermal energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Choose an equation or diagram for energy transfer. Explain when it stops working.

SUCCESS CRITERIA - I can...

1. Distinguish warming, phase change and work.
2. Use correct signs and units.
3. Explain how a predicted result depends on what you treat as true.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “phase change” and “internal energy” to explain your answer. Sentence starters are optional.

phase change: a change between states such as solid and liquid

internal energy: energy stored in a system’s microscopic motion/interactions

efficiency: useful output divided by input

01 Warm-up: bring back an earlier idea

A heat engine receives **500 J** heat and outputs **150 J** work per cycle. **Find efficiency** and rejected heat.

Word help: cycle = a process that returns a system to its initial state

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $Q = mc\,\Delta T$; $Q = mL$; $\Delta U = Q - W_{by}$; $\eta = W_{out}/Q_{hot}$. **Choose an equation** that fits and use the supplied constants. **State how you handle heat** loss, phase changes, and a full cycle.

Worked reasoning: A system takes **120 J** of heat, does **50 J** of work and increases internal energy by **70 J**. An energy account must include transfers and storage rather than treating all heat as useful work.

Your turn: Draw its system boundary, input, output and storage change.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A hot block and cool water are isolated together. **Draw initial heat flow** and **describe equilibrium** without inventing a final temperature.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Hot block

Cool water

Each outline represents the labeled sample or container.

Marzano check - circle: ① ② ③

04 Apply the physics

Compare two heating simulations. Sample A has mass **2 kg** and c = **100 J**/(kg·°C); it warms by **4°C**. Sample B has mass **2 kg** and latent heat L = **300 J/kg**; it completes a phase change at constant temperature. **Label each process** and **find its energy input**.

Sentence starter: Both ... . They differ in ... because ... .

Sample	Process	Energy equation	Energy (J)
A	Warming	____	____
B	Phase change	____	____

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

An engine absorbs **500 J** and produces **150 J** work in a cycle. **Find rejected heat** and **explain why an energy balance** alone does not establish that any temperature pair can achieve that efficiency.

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A device transfers **24 J** in **4 s**, then performs the same transfer in **8 s**. **Find both average powers** and **explain the difference**.
Distinguish transferred energy from its transfer rate.

Sentence starter: Both … . They differ in … because … .

Case	Energy (J)	Time (s)	Power (W)
A	____	____	____
B	____	____	____

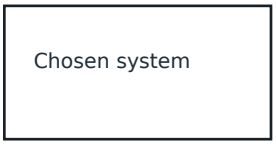
Marzano check - circle: ① ② ③

07 Exit check

A system absorbs **40 J** and does **15 J** work.
1. Find ΔU using the unit convention.
2. **Label energy entering**, leaving or stored on the diagram of the objects you are studying.

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

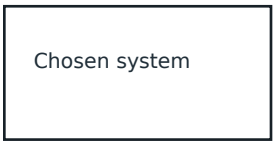
Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

1. Design a second process with the same **70 J** stored increase but a different work output.
2. **Explain why internal-energy change alone** cannot find heat input.
3. **Label energy entering**, leaving or stored on the diagram of the objects you are studying.

Sentence starter: Both … . They differ in … because … .

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: ① ② ③